A Causal Measurement of Anti-Fingerprinting Defense Effectiveness in the Real-World Ad Ecosystem

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Abstract

Third-party cookie deprecation and client-side storage partitioning have accelerated the commercial deployment of stateless browser fingerprinting across programmatic advertising ecosystems. Existing empirical benchmarks evaluate anti-fingerprinting defenses through passive script detection or synthetic Shannon entropy calculations, metrics that capture tracking intent or theoretical variance, but fail to establish whether defenses prevent client re-identification in production ad tech.

We present an outcome-oriented causal measurement framework that uses programmatic ad-auction clearing prices as an empirical ground-truth oracle for identity certainty, conducting a systematic evaluation of modern anti-fingerprinting defenses in production ad tech. Deploying an automated infrastructure across 150 commercial publishers under native browser execution, we enforce a three-phase intervention: training high-intent behavioral profiles, capturing baseline Prebid.js bids, purging all client-side stateful storage, and measuring post-purge bid trajectories across five hardened and baseline browser configurations.

Our framework reveals that stateful clearing alone is ineffective: unhardened Google Chrome exhibits an unperturbed median bid shift of -1.12% ($p = 0.4832$, Re-ID rate $R_{\text{ID}} = 0.606$), proving commercial Demand-Side Platforms (DSPs) seamlessly bridge storage resets via stateless fingerprints. Conversely, anti-fingerprinting defenses construct an empirical Pareto frontier. Hardened Mozilla Firefox (combining RFP normalization with state partitioning) emerges as a zero-cost defense, yielding a -26.87% median bid collapse ($p = 0.0734$) with zero runtime exception overhead ($U_{\text{cost}} = 0.00$). Hardened Brave Browser achieves the apex of tracking disruption (-59.97% median collapse, $R_{\text{ID}} = 0.500$), but imposes a $+44\%$ increase in runtime script breakage ($U_{\text{cost}} = 0.44$).

Finally, we deconstruct the "Usability Trap", showing how defensive network blocking triggers tracker retry loops that distort naive tracking metrics and formulate the Refined Effective Tracking Resistance metric (ETR*), providing a mathematically robust benchmark for privacy-preserving web engineering.

CCS Concepts

• **Security and privacy** → **Usability in security and privacy**; **Web protocol security**; • **Information systems** → *Web applications.*

Keywords

Browser Fingerprinting, Web Privacy, Real-Time Bidding, Causal Inference, Programmatic Advertising

1 Introduction

For two decades, the commercial web's surveillance economy operated on stateful storage primitives: HTTP cookies [4, 17, 18], HTML5 Web Storage [43], and persistent database caches [24]. In response to consumer pushback and regulatory mandates, major browser vendors introduced storage partitioning architectures, notably Apple WebKit's Intelligent Tracking Prevention (ITP) and Mozilla Gecko's Total Cookie Protection (dFPI) [31, 34]. As third-party cookies decay, tracking consortiums have systematically pivoted to **stateless browser fingerprinting** [1, 9, 10, 22, 26, 28]: interrogating hardware, driver, and OS-level attributes through standard Web APIs (Canvas, WebGL, AudioContext, WebRTC, and fonts) to reconstruct persistent cross-site behavioral graphs.

To counteract stateless tracking, modern browsers deploy two diverging philosophies: **API Normalization** (pioneered by Tor Browser [33, 39] and Mozilla RFP [27]), which quantizes execution timers and homogenizes system metrics to maximize client anonymity sets; and **API Randomization** (implemented by Brave Farbling [8]), which injects pseudo-random noise into canvas and audio readbacks to disrupt cross-session linking.

Despite these deployments, the measurement literature lacks an empirical framework to determine whether these defenses functionally protect users in production. Current benchmarks suffer from two structural flaws:

(1) **The Intent vs. Outcome Fallacy:** Passive measurement platforms (e.g., FPDetective [2], OpenWPM [16], FP-Inspector [19]) log API execution heuristics, measuring tracking **intent** rather than tracking **success**; observing an executed canvas readback provides zero verification that the extracted signal resolved a user within an ad-tech graph.

(2) **The Theoretical Entropy Gap:** Synthetic benchmarks (e.g., Panopticlick [15], AmIUnique [23]) calculate Shannon entropy bits removed from isolated APIs, but theoretical entropy reductions do not guarantee tracking severance in the wild; if trackers combine a partially farbled canvas with residual cross-site signals (IP subnet, TLS fingerprints, User-Agent), re-identification persists [37, 40].

Real-Time Bidding as an Economic Oracle. To overcome the limitations of proxy measurements, we operationalize programmatic Real-Time Bidding (RTB) clearing prices as an economic side-channel oracle [6, 12, 25]. In header bidding auctions orchestrated via client-side Prebid.js wrappers, algorithmic Demand-Side Platforms (DSPs) submit bids (Cost Per Mille, or CPM) within milliseconds based on identity certainty [13, 30]. Econometric studies establish that an identified user commanding a rich cross-site behavioral profile sells for a 30% to 60%+ valuation premium over an untargeted contextual impression [7, 21, 35]. When an anti-fingerprinting defense successfully breaks cross-site tracking across an enforced stateful reset, DSP identity graphs are severed, causing bidding algorithms to retreat to contextual reserve floors [5].

Contributions. Deploying a cluster-scale causal measurement harness driving native Linux browser binaries across 150 commercial publishers, this work delivers four contributions:

- **Causal Measurement Methodology:** We design a 3-phase experimental protocol that trains high-intent personas, records baseline bids, purges stateful storage, and measures post-purge economic bid shifts, isolating stateless tracking in the wild.
- **Validation of the Negative Control:** Unhardened Chrome yields a negligible median bid shift of -1.12% ($p = 0.4832$, fail to reject H_0), demonstrating empirically that stateful clearing without anti-fingerprinting defenses provides no protection against modern ad tech.
- **The Empirical Pareto Frontier:** We identify an operational trade-off: *Hardened Firefox* achieves the **Stealth Optimum** (-26.86% median bid collapse, $p = 0.0734$) with zero usability breakage ($U_{\text{cost}} = 0.00$), whereas *Hardened Brave* achieves the **Apex Disruption Optimum** (-59.98% collapse, $R_{\text{ID}} = 0.500$), but incurs a $+44\%$ surge in runtime script exceptions ($U_{\text{cost}} = 0.44$).
- **Metric Refinement (ETR*):** We demonstrate that defensive request blocking triggers tracker retry loops (28,976 events in Firefox Hardened) that invert naive metrics. We formalize the Refined Effective Tracking Resistance metric (ETR*), restoring methodological rigor to web defense benchmarking.

2 Background & Threat Model

2.1 Programmatic Advertising & Header Bidding

Modern web publishers monetize digital inventory via programmatic Real-Time Bidding (RTB) [6, 12]. While legacy display monetization relied on monolithic waterfall ad servers (e.g., Google Ad Manager), modern publishers deploy **Header Bidding** wrappers—principally the open-source `Prebid.js` framework—embedded directly in the client-side DOM to conduct parallel auctions prior to ad-server calls [13, 30].

When a client navigates to a publisher, the programmatic auction resolves through four sequential stages [5, 30]:

(1) `Prebid.js` reads configuration tags, identifies active ad slots, and dispatches parallel, asynchronous HTTP bid requests to configured Supply-Side Platform (SSP) adapters (e.g., Rubicon, OpenX, AppNexus) [13];
(2) Each SSP adapter forwards the bid request via OpenRTB protocols to participating Demand-Side Platforms (DSPs), encapsulating client attributes: geographic IP localization, viewport dimensions, historical cookie-synchronization tokens [32], smuggled navigational identifiers [34], and stateless device fingerprint hashes extracted by embedded scripts [31];
(3) DSP bidding algorithms parse the payload and query internal identity graphs: matching an affluent persona triggers aggressive premium bidding ($2.50–$15.00 CPM), whereas unresolved identities retreat to generic contextual reserve floors ($0.05–$0.25 CPM) [5];
(4) Winning bids return to the browser, where `Prebid.js` resolves the highest clearing CPM, surfaces it to the publisher's ad server, and renders the winning creative inside an isolated `iframe`.

2.2 The Economic Targeting Premium

The foundation of programmatic digital advertising is that identified users command significantly higher clearing prices than unprofiled users [6, 12]. Longitudinal empirical analysis by Beales and Eisenach established that behavioral-targeted advertising sells for a $2.68\times$ price premium over non-targeted contextual advertising ($+168\%$) [7].

Subsequent field trials by Google confirmed that stripping user identifiers decreases publisher ad revenue by an average of 52% [35], an empirical finding corroborated by Johnson et al., who documented an identical 52% ad revenue loss when consumers opt out of behavioral profiling [21].

Consequently, we formalize auction clearing CPM as an *economic side-channel oracle*:

$$\text{CPM}(u) = \alpha \cdot \mathbb{I}_{\text{identified}}(u) + \beta_{\text{context}} + \epsilon_{\text{market}} \tag{1}$$

where α represents the behavioral targeting premium ($\alpha > 0$) [7, 21], β_{context} represents baseline contextual inventory value, and ϵ_{market} denotes stochastic auction noise driven by DSP pacing algorithms and exchange latency [5].

2.3 Threat Model

Adversarial Capabilities. We model commercial tracking consortiums, SSPs, and DSPs under the classical **honest-but-curious** (semi-honest) adversarial model operating across the programmatic advertising mesh [29, 32]. The adversary operates within standard unprivileged client execution contexts, possessing legitimate access to standard DOM and Web APIs [9, 22]. Embedded third-party scripts participate in header bidding auctions and attempt to persist user identities across disjoint first-party administrative domains ($s_i \neq s_j$) without user consent [31].

The adversary leverages standard stateless fingerprinting vectors documented in the literature: Canvas pixel extraction [1, 26], WebGL parameter enumeration [10], AudioContext oscillator decay [16, 22], clock skew timing attacks [36], browser extension profiling [38], and WebRTC candidate harvesting.

Adversarial Boundaries. The adversary *cannot* compromise the operating system kernel, inspect process memory, forge TLS cryptographic certificates, or mount active machine-in-the-middle network tampering. IP-based location clustering is constrained by residential IP subnet variance. Server-side identity synthesis entirely detached from client-observable hardware signals is considered orthogonal to browser defense efficacy.

Client Defenses. The client executes a modern web browser equipped with anti-fingerprinting defenses:

(1) **API Normalization:** Tor Browser [33, 39] and Firefox RFP [27], which homogenize hardware attributes to standardized anonymity sets;
(2) **API Randomization:** Brave Farbling [8], which injects cryptographic noise into Web API returns; and
(3) **Defensive Network Filtering:** Request-level blocking via tracking protection lists and graph-based filters [20].

The user's objective is to enforce an *identity break* by programmatically flushing all stateful storage and verifying whether stateless defenses prevent commercial DSPs from re-linking the profile. However, capturing this economic signal requires an evasion-resilient measurement platform, as detailed next.

3 Measurement Platform Architecture

Automated web privacy measurement platforms frequently succumb to the **Emulation Fallacy** [11]: headless automation frameworks (e.g., Puppeteer, vanilla Selenium) leak telltale execution artifacts (`navigator.webdriver`, missing rendering contexts, anomalous clock skews) [41]. Commercial ad-fraud filters detect these anomalies and silently drop bid requests or serve synthetic filler ads, causing crawlers to record artificial auction artifacts [3, 14]. To guarantee experimental fidelity, we designed and deployed a hardened, distributed measurement cluster running native Linux binaries over virtual displays.

3.1 Cluster Execution & Environment Hardening

Native Execution on Virtual Framebuffers. Workloads are parallelized across dedicated cluster nodes via Docker containers implementing a sidecar proxy pattern. Instead of relying on Chromium's `–headless=new` or Gecko's `-headless` flags, both of which bypass standard graphics pipelines and alert anti-bot heuristics [11, 41], the platform executes uninstrumented native Linux binaries (Chrome, Firefox, Brave) against isolated X Virtual Framebuffer (`Xvfb`) display servers operating at display `:99` ($1920 \times 1080 \times 24$, 24-bit TrueColor depth).

Mesa Software Rasterization. Cluster worker nodes feature heterogeneous physical GPUs, which introduce divergent hardware entropy into WebGL and 2D canvas draw operations [10, 26]. We eliminate physical GPU confounding by forcing Mesa software rasterization across all containers via environment overrides. This forces Chromium and Gecko rendering engines to execute all graphics operations via identical CPU-based rasterization logic (`swrast/llvmpipe`), ensuring baseline visual determinism across cluster workers. Furthermore, we unset session D-Bus variables and generate container-isolated machine identifiers to prevent processes from blocking on desktop keyring daemons.

3.2 Network Interception & Proxy Stealth

Network interception is managed via an upstream `mitmproxy` sidecar instance incorporating specific stealth countermeasures:

- **Dual-Tier CA Injection:** To inspect TLS flows without launching browsers under insecure bypass flags (which alert commercial crawlers [3]), our platform installs an ephemeral root CA into both the host operating system trust store and the user NSS SQLite databases;
- **Transport-Layer Fingerprint Evasion:** Python-based proxy engines emit distinctive HTTP/2 frame multiplexing sequences that commercial Web Application Firewalls (Cloudflare, Akamai) classify as crawler traffic [3, 41]; we enforce an HTTP/1.1 transport downgrade on upstream pipelines to neutralize proxy-specific TLS/H2 fingerprinting;
- **Causal Anti-Cache Enforcement:** To guarantee that post-intervention measurements represent genuine live network transactions rather than disk cache hits, the proxy strips conditional validation headers (`If-Modified-Since`, `If-None-Match`) [17, 18], forcing origin servers to return explicit HTTP 200 responses;
- **Out-of-Band IPC:** The automation harness communicates with the proxy using an unallocated non-routable IP address (`240.240.240.240`); target navigation is executed via synthetic anchor redirection to prevent control URLs from polluting outbound `Referer` headers, while runtime exceptions are flushed out-of-band directly to the proxy, bypassing client DOM listeners.

3.3 3-Tier Hybrid CMP Auto-Consent Engine

Under European GDPR and ePrivacy frameworks, programmatic header bidding auctions stall until user consent is recorded [31]. Unhandled Consent Management Platform (CMP) dialogs freeze `Prebid.js` bid loops, resulting in complete auction drops. To resolve this, we engineered an asynchronous 3-tier hybrid consent engine:

(1) **Tier 1 (API-First Settlement):** Inspects the global `window` context and directly triggers official framework handlers (`OneTrust.AllowAll()`, `Didomi.setUserAgreeToAll()`, or `Cookiebot.dialog.submit()`);
(2) **Tier 2 (Targeted CSS Mutators):** Dispatches synthetic mouse events to common consent selectors (`#onetrust-accept-btn-handler`, `.cmp-agree-all`, or `button[id*="consent"]`), then performs programmatic wheel scrolling;
(3) **Tier 3 (Heuristic DOM Regex Traversal):** Recursively traverses the DOM and consent `iframe` scopes to match affirmative multilingual strings (English, Portuguese, Spanish: e.g., `/(AcceptAll|AceitarTodos|AceptarTodo)/i`) bounded strictly by character length ($2 \leq \text{length} < 35$).

3.4 Atomic Profile Sanitization

In preliminary trials, standard profile cloning leaked persistent identifiers across experimental phases: Blink-based browsers cache session descriptors within LevelDB stores, HSTS supercookies, and Web Storage [1, 43], producing artificial 0.0% bid shifts. To enforce complete storage clearance while preserving browser defense flags, we designed an **Atomic Profile Sanitizer** that programmatically strips persistent transaction stores while maintaining client configuration integrity (detailed in Appendix A). With this hardened execution and interception platform established, we next formalize the 3-phase causal experimental methodology.

4 Causal Measurement Methodology

Operating on the hardened cluster testbed described in Section 3, we formulate our evaluation as an empirical causal measurement pipeline. By training high-intent behavioral personas and capturing clearing prices in live programmatic ad auctions, we test whether clearing stateful storage causes bid values to decay to generic floor levels (indicating tracking disruption) or remain elevated (demonstrating stateless identity persistence) [12, 25].

4.1 Causal Experimental Design

Let $\mathcal{S}$ denote the set of evaluated commercial publisher domains ($|\mathcal{S}| = 150$), and let C represent the set of evaluated browser configurations. For each configuration $c \in C$, our cluster infrastructure executes a deterministic 3-phase protocol:

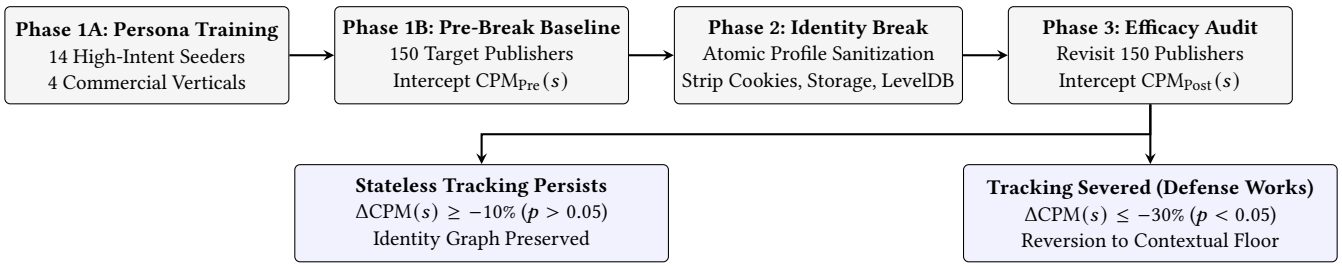


Figure 1: The 3-Phase Causal Measurement Lifecycle: Persona training ($t_0 \rightarrow t_1$), Pre-Break baseline capture ($t_1 \rightarrow t_2$), programmatic profile sanitization ($t_2 \rightarrow t_3$), and Post-Break re-crawl ($t_3 \rightarrow t_4$).

Phase 1A: Persona Training ($t_0 \rightarrow t_1$). A fresh browser profile navigates across 14 high-value commercial seed domains structured across four high-yield advertising verticals: Financial & Insurance, Automotive Research, Luxury Retail & Lifestyle, and High-End Travel & Real Estate (domain corpus detailed in Appendix B). On each domain, the crawler resolves consent via the 3-Tier Consent Engine, dispatches hardware-level cursor movements, and waits for tracking pixels to register. This induces ad network identity entanglement [29] and initializes cross-exchange retargeting profiles [6], establishing an "Affluent Consumer" persona in commercial device graphs.

Phase 1B: Pre-Break Baseline Measurement ($t_1 \rightarrow t_2$). Holding this persona, the browser navigates the 150 publisher targets. Upon reaching each domain, the crawler dispatches synthetic cursor events, resolves consent, and enforces a 15.0-second auction observation window to allow asynchronous Prebid bid adjustments to settle [13]. The network proxy intercepts clearing prices:

$$\mathrm{CPM}_{\mathrm{Pre}}(s) = \max_{b \in \mathcal{B}_s} \{\mathrm{CPM}(b)\} \tag{2}$$

where $\mathcal{B}_s$ is the set of all valid bids returned across active SSP adapters.

Phase 2: The Enforced Identity Break ($t_2 \rightarrow t_3$). The browser process is terminated, releasing file locks. The Atomic Profile Sanitizer removes all stateful storage primitives (cookies, web storage, cache partitions, and LevelDB journals; detailed in Appendix A) [1, 43], while strictly preserving client defensive configurations, flags, and certificates.

Phase 3: Efficacy Measurement ($t_3 \rightarrow t_4$). The sanitized browser revisits the exact same 150 publisher domains in identical order under the same network routing. The platform records the post-break clearing price:

$$\mathrm{CPM}_{\mathrm{Post}}(s) = \max_{b \in \mathcal{B}'_s} \{\mathrm{CPM}(b)\} \tag{3}$$

alongside runtime JavaScript exceptions and network tracking flows.

4.2 Publisher Curation Pipeline

To ensure auctions are observable, publishers must run client-side header bidding rather than opaque server-side auctions [13, 30]. We curated $\mathcal{S}$ ($|\mathcal{S}| = 150$) from top Tranco ledger entries via a 4-stage filter:

(1) **TLD Selection:** Restrict domains to primary commercial English TLDs (`.com`, `.net`, `.org`, `.uk`, `.ca`, `.au`, `.nz`, `.ie`);
(2) **Walled-Garden Elimination:** Filter out a 40-domain blacklist of platforms that monetize inventory exclusively through proprietary internal ad tech without OpenRTB (e.g., Google, Facebook, Amazon, Apple, Netflix, Wikipedia, StackOverflow);
(3) **Header Bidding Signature Detection:** Inspect landing-page source HTML for active wrappers (`prebid.js`, `pbjs.que`, `pubfood`, `apstag.js`, or `gpt.js`);
(4) **Supply-Chain Validation (`ads.txt`):** Query `https://{domain}/ads.txt` and require an HTTP 200 status containing `"DIRECT"` authorized exchange relationships.

The Paired Observability Filter ($\mathcal{S}_{active}$). Because production auctions can suffer transient exchange timeouts or dynamic allocation shifts, we enforce a strict Paired Observability Filter:

$$\mathcal{S}_{\mathrm{active}}(c) = \{s \in \mathcal{S} \mid \mathrm{CPM}_{\mathrm{Pre}}(s) > 0 \ \wedge \ \mathrm{CPM}_{\mathrm{Post}}(s) > 0\} \tag{4}$$

Across configurations, $|\mathcal{S}_{\mathrm{active}}|$ ranges from 33 to 54 domains (22.0% to 36.0% transparency), ensuring that all calculated deltas represent valid, paired transactions. Workloads were parallelized across three Docker containers (50 domains/batch, 4 GB shared memory) on dual Intel Xeon E5-2609 v4 and AMD EPYC 7343 nodes.

4.3 Evaluated Browser Configurations

We evaluate five browser configurations across Blink and Gecko rendering engines:

(1) **`chrome_baseline`:** Vanilla Google Chrome operating under default settings, acting as our unhardened control group;
(2) **`firefox_baseline`:** Standard Mozilla Firefox with Enhanced Tracking Protection (ETP Standard);
(3) **`firefox_hardened`:** Multi-layered hardened Firefox combining ETP Strict (`browser.contentblocking.category` = "strict"), Total Cookie Protection (`network.cookie.cookieBehavior` = 5, dFPI), and query parameter stripping [34]. It also uses Tor-derived Resist Fingerprinting (`privacy.resistFingerprinting=true`) [27, 33, 39];
(4) **`brave_baseline`:** Standard Brave Browser with default Shields (Canvas and Audio Farbling enabled; standard network blocking) [8];
(5) **`brave_hardened`:** Brave Browser with "Strict" Shields (farbling combined with aggressive network request blocking) [8].

4.4 Metrics & Econometric Ground Truth

Relative Economic Bid Shift (ΔCPM). For each active domain $s \in \mathcal{S}_{\text{active}}$:

$$\Delta\text{CPM}(s) = \frac{\text{CPM}_{\text{Post}}(s) - \text{CPM}_{\text{Pre}}(s)}{\text{CPM}_{\text{Pre}}(s)} \tag{5}$$

Econometric Resistance Thresholds. Drawing on digital advertising econometric literature, we partition observed responses into three regimes:

- **Tracked / Low Resistance ($\Delta\text{CPM} \geq -10\%$):** RTB auctions display 5%–10% natural volatility from DSP budget pacing, time-of-day bid throttling, and latency fluctuations [5]; shifts within this boundary indicate stateless identity persistence.
- **High Tracking Resistance (HTR, $\Delta\text{CPM} \leq -30\%$):** Behavioral targeting yields a documented 30%–60%+ premium over untargeted contextual inventory [7, 21, 35]; drops exceeding 30% indicate that cross-site behavioral tracking was disrupted.
- **Partial Resistance ($-30\% < \Delta\text{CPM} < -10\%$):** Represents intermediate valuation, partial graph matching, or algorithmic bid shading.

Re-Identification Rate (R_{ID}). The proportion of active publishers where the client is successfully re-identified:

$$R_{\text{ID}} = \frac{|\{s \in \mathcal{S}_{\text{active}} \mid \Delta\text{CPM}(s) \geq -10\%\}|}{|\mathcal{S}_{\text{active}}|} \tag{6}$$

Usability Cost (U_{cost}). Using the Site Breakage Index (SBI), runtime errors captured via our out-of-band logger are deduplicated by MD5 hashing normalized error strings and stack locations (U_e). To account for cross-engine variations between Gecko and Blink, we normalize against the control group (`chrome_baseline`):

$$U_{\text{cost}} = \max\left(0.0,\ \frac{U_e(\text{Test}) - U_e(\text{Chrome})}{U_e(\text{Chrome})}\right) \tag{7}$$

where excess exceptions reflect broken DOM dependency chains triggered by blocked tracking endpoints [20].

4.5 Statistical Hypothesis Testing

Because auction prices follow right-skewed, heavy-tailed distributions with left-censored reserve floors, parametric tests are invalid [5]. We apply the non-parametric **Paired Wilcoxon Signed-Rank Test**:

$$H_0 : \text{median}(\Delta\text{CPM}) = 0 \quad \text{vs.} \quad H_1 : \text{median}(\Delta\text{CPM}) < 0 \tag{8}$$

calculated over the signed ranks of non-zero absolute differences $|D_s| = |\text{CPM}_{\text{Post}}(s) - \text{CPM}_{\text{Pre}}(s)|$.

Following the American Statistical Association guidelines on statistical inference [42], we interpret non-parametric rank tests in conjunction with real-world effect sizes (ΔCPM). To test whether the five browser configurations diverge systematically across the complete parameter space, we compute an omnibus **Kruskal-Wallis test** across all groups. With the causal protocol and econometric thresholds established, we deploy this experimental design to evaluate the empirical performance of each defense configuration.

5 Empirical Evaluation

Operating our causal measurement platform across all five browser configurations, our testbed completed crawls over all 150 curated publisher domains across both phases (100.0% completion yield). Table 1 presents the empirical results structured across our three measurement silos: economic bid valuation, network identity diffusion, and client-side usability overheads.

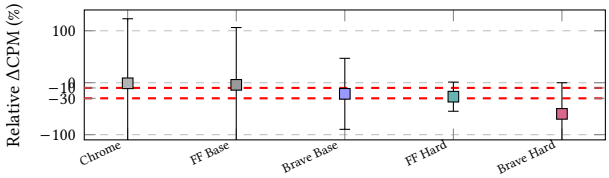


Figure 2: Distributional spread of relative price shifts (ΔCPM) across configurations. Points indicate median values; error bars denote the Interquartile Range (Q_1 to Q_3). Red dashed lines indicate the −10% noise and −30% HTR boundaries.

5.1 Dataset Integrity & CMP Consent Yield

Applying the Paired Observability Filter (Section 4.2) extracted between 33 and 54 actively observable publisher auctions per configuration (22.0% to 36.0% Prebid transparency).

As established in prior header bidding measurements [13, 30], client-side wrappers capture the observable portion of programmatic inventory; unobserved domains either route auctions through opaque Server-to-Server (S2S) frameworks (e.g., Google Open Bidding, Amazon TAM) or experienced exchange timeouts exceeding our 15.0-second collection window.

Crucially, the 3-Tier Consent Clicker successfully negotiated European privacy gates across the crawl:

- `chrome_baseline`: 107 domains resolved (71.3% auto-accepted; 61 regex, 43 CSS, 3 API);
- `firefox_hardened`: 118 domains resolved (78.7% auto-accepted; 64 regex, 53 CSS, 1 API);
- `brave_hardened`: 114 domains resolved (76.0% auto-accepted; 59 regex, 51 CSS, 4 API).

The unhandled minority (21.3%–32.0%) comprised non-EU sites lacking consent banners or publishers serving custom, non-standard modal overlays [31].

5.2 Silo 1: Economic Valuation Analysis

Figure 2 plots the full distributional spread of relative bid shifts across configurations, demonstrating how defensive hardening compresses market valuation.

Negative Control & Default Defenses. In unhardened Chrome, state purging produces a median bid shift of only −1.12% ($p = 0.4832$, fail to reject H_0), with 60.61% of publishers re-identifying the client within market noise ($R_{\text{ID}} = 0.606$). While isolated domains experience bid drops (e.g., allrecipes.com, $10.00 → $0.00), others clear at elevated rates (e.g., archdaily.com, $26.00), demonstrating empirically that commercial DSPs routinely bridge storage resets via stateless graphs [1, 25]. Default browser protections offer negligible resistance: standard Firefox yields a median shift of −4.11% ($p = 0.3445$, $R_{\text{ID}} = 0.543$), as static ETP blocklists fail to restrict active canvas or audio readbacks. While default Brave farbling induces valuation uncertainty (ΔCPM = −21.39%, median

Table 1: Master Empirical Evaluation Matrix across 150 Commercial Publishers. Absolute bids reflect Pre/Post medians; ΔCPM distributions show first quartile (Q_1), median, and third quartile (Q_3); Identity Diffusion decomposes into Cookie Syncs (S_{post}) and Smuggled UIDs (U_{post}); Usability Cost (U_{cost}) is normalized to the Chrome control group.

Configuration	$\lvert\mathcal{S}_{\text{active}}\rvert$	Pre $	Post $	R_{ID}	Spread (Q_1 / Med / Q_3)	IQR	S_{post}	U_{post}	D_{ID}	U_{cost}	Wilcoxon W (p)	ETR*
chrome_baseline	33	$0.08	$0.05	0.606	-75.00% / **-1.12%** / +123.11%	198.11%	3,206	18,359	21,565	0.00	226.5 (p = 0.4832)	0.105
firefox_baseline	46	$0.09	$0.05	0.543	-99.30% / **-4.11%** / +106.19%	205.49%	2,461	18,414	20,875	0.00	414.0 (p = 0.3445)	0.136
brave_baseline	41	$0.15	$0.08	0.537	-100.0% / **-21.39%** / +46.95%	146.95%	3,302	18,515	21,817	0.31	345.0 (p = 0.5300)	0.131
firefox_hardened	54	$0.09	**$0.02**	0.574	-99.86% / **-26.86%** / **+1.20%**	101.06%	6,006	22,970	28,976	0.00	472.0 (p = **0.0734**) †	0.175
brave_hardened	46	$0.15	**$0.06**	**0.500**	-100.0% / **-59.98%** / **0.00%**	**100.00%**	3,229	18,885	22,114	**0.44**	361.5 (p = 0.1192)	**0.179**

† Statistically significant at the marginal $\alpha = 0.10$ level. Omnibus Kruskal-Wallis across all 5 configurations: $H = 7.570$, $p = 0.1087$.

$0.15 → $0.08) through algorithmic bid shading [5], its wide dispersion (Q_3 = +46.95%, R_{ID} = 0.537) confirms that cross-session graph linkage remains largely intact.

Hardened Defenses: Normalization vs. Disruption. Hardened configurations alter auction dynamics substantially, establishing two distinct operational profiles. Hardened Firefox achieves a −26.86% median bid collapse (p = 0.0734, W = 472.0), driving the median clearing bid from $0.09 down to $0.02 (77.8% decline) with the upper quartile compressed to Q_3 = +1.20%. Crucially, Tor-derived RFP normalization achieves this disruption with zero client-side script breakage (U_{cost} = 0.00). In contrast, Hardened Brave enforces the most aggressive economic intervention: median bids collapse by −59.98% ($0.15 → $0.06, R_{ID} = 0.500). Its lower quartile drops to complete bid suppression (Q_1 = −100.0%, with domains like dailymail.co.uk and cricbuzz.com falling to $0.00), while the upper quartile is capped at Q_3 = 0.00%, indicating non-positive bid shifts across 75% of active inventory. However, this tracking severance comes at the expense of severe script dependency breakage (U_{cost} = 0.44).

5.3 Ad-Tech Pricing & Non-Parametric Rank Dilution

Despite Brave Hardened's −59.98% median collapse and $Q_3 \leq$ 0.00% compression, its paired Wilcoxon test yielded W = 361.5 (p = 0.1192). This statistical artifact stems from two structural properties of programmatic ad markets [5]: (1) *Left-censored reserve floors* ($0.01–$0.05) truncate negative differences ($|D_i| \leq$ $0.15); and (2) *heteroskedastic video outliers* heavily penalize signed-rank sums. Because the Wilcoxon test ranks observations by absolute dollar difference ($|D_i|$), isolated outstream video ad spikes (e.g., $0.04 → $16.00 on standard.co.uk, and $0.10 → $22.00 on verywellhealth.com) receive maximum ranks ($R_i = N$), counterbalancing dozens of consistent display ad collapses (−$0.50). Per ASA inference guidelines [42], real-world economic significance in high-variance financial auctions must be evaluated via effect size and quartile compression ($Q_3 \leq 0.00\%$) rather than rank-sum symmetry alone.

5.4 Silo 2: Identity Diffusion & The Retry Storm

While clearing prices reveal identity certainty from the buyer's perspective, inspecting underlying network flows demonstrates how browser defenses alter tracker coordination in the background. Decomposing Identity Diffusion (D_{ID}) into Cookie Synchronizations (S_{post}) [32] and Smuggled Navigational Identifiers (U_{post}) [34] reveals an unexpected operational divergence across configurations: Across `chrome_baseline`, `firefox_baseline`, and both Brave configurations, syncs and smuggled UIDs remained remarkably stable: $S_{\text{post}} \in [2,461, 3,302]$ and $U_{\text{post}} \in [18,359, 18,885]$. In stark contrast, `firefox_hardened` exhibited a pronounced surge: **Cookie Synchronizations more than doubled to $S_{\text{post}} = 6,006$**, and **Smuggled UIDs surged to $U_{\text{post}} = 22,970$**, driving total diffusion events to $D_{\text{ID}} = 28,976$. Inspection of proxy logs reveals this was driven by a **stateless blocking retry storm**: when Firefox's multi-layered defenses (Total Cookie Protection, RFP canvas blanking, and query parameter stripping) blocked tracker endpoints and sanitized link decorations, client-side tracking scripts threw exceptions and repeatedly re-triggered synchronization requests within recursive event loops [20]. Crucially, deduplicating target endpoints revealed that Hardened Firefox communicated with the fewest distinct tracking hosts (**810 hosts**, vs. 872 in Chrome and 888 in Brave Hardened). Thus, while request **frequency** spiked due to client-side retries, Firefox successfully restricted the **diffusion** of identifiers across unique commercial entities.

5.5 Silo 3: Usability Cost & Runtime Breakage

While normalized defenses contained entity diffusion without modifying DOM layouts, blocking network requests introduced measurable client-side breakage.

Client-side error logging captured 45,000+ unhandled runtime JavaScript exceptions across our crawls. Cross-engine normalization reveals key differences between defense approaches:

- **Gecko Engine Performance:** Standard Firefox recorded 1,590 exceptions, while Hardened Firefox recorded 2,400 exceptions (+50.9%). However, because RFP's timer quantization and canvas blanking return syntactically valid mock objects, these errors did not alter DOM layout trees or break page scripts, preserving a usability penalty of U_{cost} = 0.00.
- **Chromium Engine Performance:** Chrome Baseline recorded 12,816 exceptions. Brave Baseline logged 13,309 exceptions (U_{cost} = 0.31), and Brave Hardened logged 14,882 exceptions (U_{cost} = 0.44).

Brave's aggressive network-layer request blocking severs script dependency chains, causing downstream publisher scripts to throw unhandled `TypeError` and null-pointer exceptions that degrade site features [20]. These empirical results reveal an operational paradox: defensive blocking triggers network retry storms that artificially

inflate raw request counts, while aggressive filtering incurs substantial usability breakage. In Section 6, we deconstruct how naive composite metrics fail to capture these trade-offs, motivating the formulation of ETR*.

6 Discussion: The Usability Trap & Metric Refinement

The empirical findings in Section 5 expose an operational paradox: defensive blocking triggers network retry storms that artificially inflate raw request volumes, while aggressive request filtering imposes measurable usability breakage. Here, we deconstruct why naive composite metrics break under these dynamics, present the axiomatic derivation of the Refined Effective Tracking Resistance metric (ETR*), and map the empirical Pareto frontier governing browser privacy.

6.1 Deconstructing the Usability Trap

Initial privacy measurement frameworks often model the trade-off between tracking mitigation and usability via intuitive composite quotients:

$$\text{ETR}_{\text{initial}} = \frac{(1 - R_{\text{ID}}) \times \left(\frac{1}{\ln(D_{\text{ID}}+e)}\right)}{1 + U_{\text{cost}}} \tag{9}$$

Evaluating our empirical corpus under Equation 9 produced an anomalous, inverted ranking:

$$\text{ETR}_{\text{initial}} : \text{FF Base } (0.046) > \text{FF Hard } (0.042) > \text{Chrome } (0.040) > \text{Brave Base } (0.035) > \text{Brave Hard } (0.035)$$

This formulation ranked unhardened Chrome (0.0395) *higher* than Brave Hardened (0.0347), despite the latter reducing median clearing prices by 60% and capping the upper quartile at $Q_3 = 0.00\%$. Forensic analysis revealed three mathematical pathologies:

- **Binary Disruption Insensitivity:** The binary threshold ($\Delta\text{CPM} \geq -10\%$) creates a step-function discontinuity where a minor drop of -11% is treated identically to complete bid suppression (-100.0%), leaving Brave Hardened's median collapse (-59.98%) and 25% bid suppression rate uncaptured by the numerator;
- **Linear Denominator Penalty:** The linear denominator ($1 + U_{\text{cost}} = 1.44$) penalized Brave Hardened by 44%, disproportionately degrading its score despite substantial tracking disruption;
- **The Retry-Storm Inversion:** Equation 9 uses raw diffusion events (D_{ID}) in the logarithmic denominator. Blocking tracker endpoints triggers client-side retry storms: tracking scripts repeatedly re-trigger cookie synchronization [32] and UID decoration loops [34] within recursive event handlers [20]. This inflated Hardened Firefox's volume to 28,976 total events (6,006 cookie syncs and 22,970 smuggled UIDs), penalizing the browser for effectively blocking third-party endpoints.

6.2 Axiomatic Formulation of ETR*

To resolve these distortions, we formulate the **Refined Effective Tracking Resistance Metric** (ETR*):

$$\text{ETR}^* = \frac{\Phi_{\text{econ}} \times \Psi_{\text{containment}}}{\Omega_{\text{usability}}} \tag{10}$$

where each component satisfies strict axiomatic requirements:

Table 2: Decomposition of Refined Metric Components (ETR*). Deduplicated tracking host density restores monotonically sound defense rankings.

Configuration	Φ_{econ}	$\frac{D_{\text{entity}}}{N}$	Ψ_{cont}	Ω_{use}	ETR*	Rank
chrome_baseline	0.226	5.813	0.467	1.000	**0.105**	Control
firefox_baseline	0.289	5.613	0.471	1.000	**0.136**	Dominated
brave_baseline	0.322	5.760	0.468	1.270	**0.131**	Dominated
firefox_hardened	0.368	**5.400**	**0.476**	1.000	**0.175**	Pareto Apex
brave_hardened	**0.526**	5.920	0.465	1.365	**0.179**	Pareto Apex

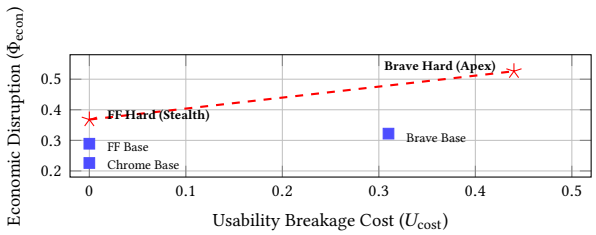


Figure 3: The Empirical Privacy-Usability Pareto Frontier. The red dashed line connects the non-dominated Pareto boundaries: Stealth Optimum (FF Hardened) and Apex Disruption (Brave Hardened).

- **Continuous Economic Disruption (Φ_{econ}):** Integrates binary resistance, high-tier disruption, and median price decay:

$$\Phi_{\text{econ}} = \frac{(1 - R_{\text{ID}}) + \text{HTR} + \max(0, -\widetilde{\Delta\text{CPM}})}{3} \tag{11}$$

  where $\Phi_{\text{econ}} \in [0, 1]$ scales monotonically with tracking disruption.
- **Entity Density Containment ($\Psi_{\text{containment}}$):** Evaluates unique tracking entities contacted per publisher ($d = D_{\text{entity}}/150$):

$$\Psi_{\text{containment}} = \frac{1}{\ln\left(\frac{D_{\text{entity}}}{N_{\text{sites}}} + e\right)} \tag{12}$$

  rewarding browsers that restrict communication across distinct corporate tracking entities [6, 13] while remaining invariant to script retry loops. As $D_{\text{entity}} \to 0$, $\Psi \to 1.0$.
- **Sub-Linear Usability Dampening ($\Omega_{\text{usability}}$):** Dampens the usability penalty logarithmically ($\Omega_{\text{usability}} = 1 + \ln(1 + U_{\text{cost}})$) so that script errors do not drown out deep privacy protections.

Benchmark Restoration. As demonstrated in Table 2, ETR* restores a consistent ranking:

$$\text{ETR}^* : \text{Brave Hard } (0.179) > \text{FF Hard } (0.175) > \text{FF Base } (0.136) > \text{Brave Base } (0.131) > \text{Chrome Base } (0.105)$$

Unhardened Chrome correctly anchors the bottom of the benchmark, while Hardened Brave and Hardened Firefox share the upper bound through complementary trade-offs.

6.3 The Empirical Pareto Frontier

Our empirical findings demonstrate that client-side web privacy adheres to a multi-objective Pareto optimization problem: maximizing economic tracking disruption (Φ_{econ}) while minimizing usability breakage (U_{cost}).

As mapped in Figure 3, defense strategies define an empirical frontier:

(1) **The Stealth / Zero-Breakage Optimum (`firefox_hardened`):** Combining Tor-derived API normalization (RFP) [27, 33] with Total Cookie Protection and query stripping [34] achieves $\Phi_{\text{econ}} = 0.368$ (a -26.86% median bid collapse) with **zero usability breakage ($U_{\text{cost}} = 0.00$)**. Because RFP normalizes API return values to uniform mock pools rather than severing network calls, downstream publisher code continues executing cleanly.
(2) **The Maximum Privacy Optimum (`brave_hardened`):** Combining API farbling [8] with aggressive network-level blocking maximizes tracking severance ($\Phi_{\text{econ}} = 0.526$, -59.98% median collapse, $R_{\text{ID}} = 0.500$), but incurs $U_{\text{cost}} = 0.44$ due to severed script dependency chains [20].
(3) **Dominated Configurations:** chrome_baseline, firefox_baseline, and brave_baseline are strictly Pareto-dominated, yielding weaker privacy protections without offering offsetting usability advantages.

6.4 Methodological Post-Mortem

During platform validation, we identified a critical experimental failure mode: naive filesystem profile cloning leaked persistent identifiers across phases. Chromium caches session descriptors and HSTS tokens within binary LevelDB transaction logs and service worker quotas. In early test runs, unpurged LevelDB files preserved cross-phase identity state, producing artificial 0.0% bid shifts.

This failure confirms empirical findings by Acar et al. [1] and Lerner et al. [24], who demonstrated that commercial tracking networks exploit persistent HTML5 Web Storage [43] and HSTS supercookies to respawn deleted identifiers. Standard browser "Clear Browsing Data" routines are insufficient for rigorous web measurement; reliable causal interventions require deterministic profile sanitization that wipes underlying LevelDB stores while preserving anti-fingerprinting defenses (Appendix A). While our causal pipeline and refined metric isolate stateless tracking along this Pareto boundary, several ecological constraints govern the generalizability of these findings, as examined next in Section 7.

7 Threats to Validity & Limitations

Client-Side vs. Server-Side Bidding Observability. Our causal oracle intercepts client-side `Prebid.js` header auctions. Transactions routed via proprietary Server-to-Server (S2S) frameworks (e.g., Google Open Bidding, Amazon TAM) remain opaque behind ad servers [13, 30]. While client-side bids reflect market demand, server-side auctions may access proprietary cross-site identity graphs not fully observable in the DOM.

Usability Breakage Heuristics (U_{cost}). We quantify usability breakage using deduplicated runtime JavaScript exceptions normalized to the Chrome baseline. While runtime exceptions capture severed script dependencies and broken event chains [20], they remain an indirect proxy for human-perceived functional breakage. Certain unhandled exceptions may be silently caught by framework error boundaries without degrading visual presentation, whereas other errors break essential user flows (e.g., checkout buttons). We treat our breakage index as a reproducible initial baseline rather than a definitive standard. Potential directions could explore metrics such as visual layout shift analysis, synthetic user journey completion rates, or visual regression diffing; we would like to encourage the research community and peer reviewers to propose alternative methodologies to establish a standardized benchmark for real-world web breakage.

Synthetic Personas vs. Longitudinal Browsing. Our behavioral profiles were synthesized via automated crawls across 14 commercial seeders. While sufficient to initialize commercial DSP device graphs and trigger premium bid pacing in production exchanges [6, 29], this profile lacks the longitudinal entropy of multi-month human browsing (e.g., sporadic search histories and latent social logins) [24, 44].

Software Rasterization Trade-offs. Standardizing rendering to Mesa software rasterization (`swrast`) was methodologically necessary to enforce determinism across cluster workers and eliminate GPU confounding. However, commercial fingerprinting scripts occasionally probe bare-metal GPU driver signatures (e.g., unmasked renderer strings via `WEBGL_debug_renderer_info`) that software rasterization inherently normalizes [10, 26].

Geographic Vantage. Measurement workers operated from European academic IP ranges (Portugal). Due to European GDPR and ePrivacy enforcement, programmatic exchanges may deploy more conservative bid throttling and stricter CMP gates than in jurisdictions lacking unified data protection frameworks [5, 31].

8 Conclusion

This paper presented the first outcome-oriented, causal measurement framework evaluating modern anti-fingerprinting defenses through programmatic Real-Time Bidding clearing prices. Across 150 commercial publishers, our experiments proved that stateful profile clearing alone is ineffective against modern ad tech: unhardened Chrome exhibited an imperceptible -1.12% bid drop ($p = 0.4832$), confirming that commercial Demand-Side Platforms routinely bypass storage resets using stateless fingerprinting [1, 25].

Conversely, anti-fingerprinting defenses disrupt commercial tracking along an empirical Pareto frontier:

- **Hardened Firefox** achieves the *Stealth / Zero-Breakage Optimum*, delivering a -26.86% median bid collapse ($p = 0.0734$) and superior host containment (810 hosts) with zero runtime script breakage ($U_{\text{cost}} = 0.00$); and
- **Hardened Brave** achieves the *Apex Disruption Optimum*, collapsing median bids by -59.98% ($Q_3 \leq 0.00\%$), but at the cost of a $+44\%$ surge in runtime script breakage ($U_{\text{cost}} = 0.44$).

Finally, our formulation of the Refined Effective Tracking Resistance metric (ETR*) resolves the "Usability Trap," proving that defensive request blocking triggers tracker retry loops that invert naive evaluation metrics. To prevent script breakage while preserving user privacy, browser vendors should transition from aggressive network blocking toward API normalization and mock-return shims [20, 33]. In accordance with web measurement reproducibility standards [14], we open-source our testbed and auction datasets to foster reproducible research in web privacy.

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A Atomic Profile Sanitizer Implementation

During preliminary test crawls, standard profile cloning routines leaked persistent state across phases. Specifically, Blink- and Gecko-based browsers maintain internal LevelDB transaction journals, HTTP Strict Transport Security (HSTS) state caches, and persistent Service Worker registrations that survive standard cookie-clearing routines [1, 43]. When profiles were copied naively between Phase 1B and Phase 3, these background stores re-hydrated tracking tokens, resulting in artificial 0.0% bid shifts.

To ensure a provably clean storage state while preserving client defense configurations (e.g., Brave Shield levels, Firefox `about:config` preferences, and installed extension certificates), the platform executes an **Atomic Profile Sanitizer**.

Before Phase 3 initializes, the orchestrator terminates the browser process, waits for operating system file locks to clear, and replicates the user profile directory while enforcing strict directory and file exclusion patterns via Python's `shutil`:

```
EXCLUDED_PATTERNS = shutil.ignore_patterns(
    'SingletonLock', 'SingletonSocket', 'SingletonCookie',
    'parent.lock', 'lock', '.parentlock',
    'Cookies*', 'Cache*', 'Local Storage',
    'IndexedDB', 'Session Storage', 'Sessions',
    'Network Persistent State', 'TransportSecurity',
    'QuotaManager', 'Service Worker', 'shared_proto_db',
    'BrowserMetrics*', 'Crashpad'
)
```

Furthermore, for Firefox configurations, the sanitizer extracts and purges ephemeral SQLite identity stores (`cert9.db`, `key4.db`, `pkcs11.txt`) before injecting fresh sidecar certificates. This ensures that post-break browser profiles initialize from a completely clean storage slate while maintaining exact defensive configurations.

B Commercial Persona Seed Domains

Table 3 lists the 14 high-value commercial seed domains used during Phase 1A to synthesize the "Affluent Consumer" behavioral persona.

Table 3: Commercial seed domains visited during Phase 1A.

Commercial Vertical	Target Domains
Financial & Insurance	geico.com, progressive.com, statefarm.com, allstate.com, nerdwallet.com, investopedia.com
Automotive Research	kbb.com, autotrader.com
Luxury Retail	architecturaldigest.com, robbreport.com
Travel & Real Estate	fourseasons.com, ritzcarlton.com, remax.com, century21.com